Pioneer PK-7

Packaging Conveyor - Synthetic Operations Handbook

ASSET REFERENCE	SYN-PK-007
DOCUMENT STATUS	Revision 1.5 / Synthetic edition
DATA CLASS	Entirely fictional / non-production

1. Purpose and equipment summary

The fictional Pioneer PK-7 moves sealed cartons through a synthetic inspection cell. It uses an encoder, entry photoeye, belt-drift switches, and independently adjustable take-up screws.

This manual exists only to exercise document upload, PDF extraction, page provenance, retrieval, and citation behavior in QIP. It does not describe real equipment and must not be used for physical maintenance.

Nominal operating envelope

Parameter	Synthetic value
Belt width	600 mm
Nominal speed	0.65 m/s
Tracking tolerance	+/- 3 mm
Tension-side mismatch limit	4 mm
Photoeye cleaning interval	Weekly
Drive current alert	6.8 A for 10 seconds

2. Alarm and troubleshooting reference

Preserve observations before resetting an alarm. The table provides decision support for synthetic investigations; it does not establish root cause by itself.

E17	**Right belt-drift switch active**

Recommended inspection: Stop the conveyor. Remove debris, measure both take-up positions, and correct mismatch only when it exceeds 4 mm. Adjust in quarter-turn increments.

E06	**Entry photoeye blocked for 12 seconds**

Recommended inspection: Remove the carton under safe access, clean the lens, and verify bracket alignment. Do not bypass the sensor.

D31	**Drive current above 6.8 A**

Recommended inspection: Inspect belt obstruction and roller freedom before changing motor parameters.

INVESTIGATION EXERCISE	Scenario PK-C: E17 occurs twice after carton debris is removed. Belt offset is 8 mm right and take-up mismatch is 6 mm. Evidence supports correcting take-up asymmetry in quarter-turn increments, then observing five unloaded revolutions before production restart.

3. Inspection and maintenance

Use the following synthetic intervals to test evidence capture, question answering, and citation to a specific page.

- At shift start: test emergency stops, guards, drift switches, and photoeye response.
- Weekly: clean the photoeye lens and measure left/right take-up positions from the fixed datum.
- Every 750 synthetic hours: inspect belt splice, roller bearings, and gearbox leakage.
- After E17: record the measured tracking offset and take-up mismatch before making an adjustment.

Safety boundary

TEST DATA ONLY	Lockout, isolation, and competent-person requirements in this fictional document are illustrative. Never apply these values or procedures to real machinery.

Suggested QIP test questions

- What observation should be preserved before changing a machine setting?
- Which threshold in the source supports the recommended first inspection?
- Is there enough evidence to state a confirmed root cause? Explain the uncertainty.